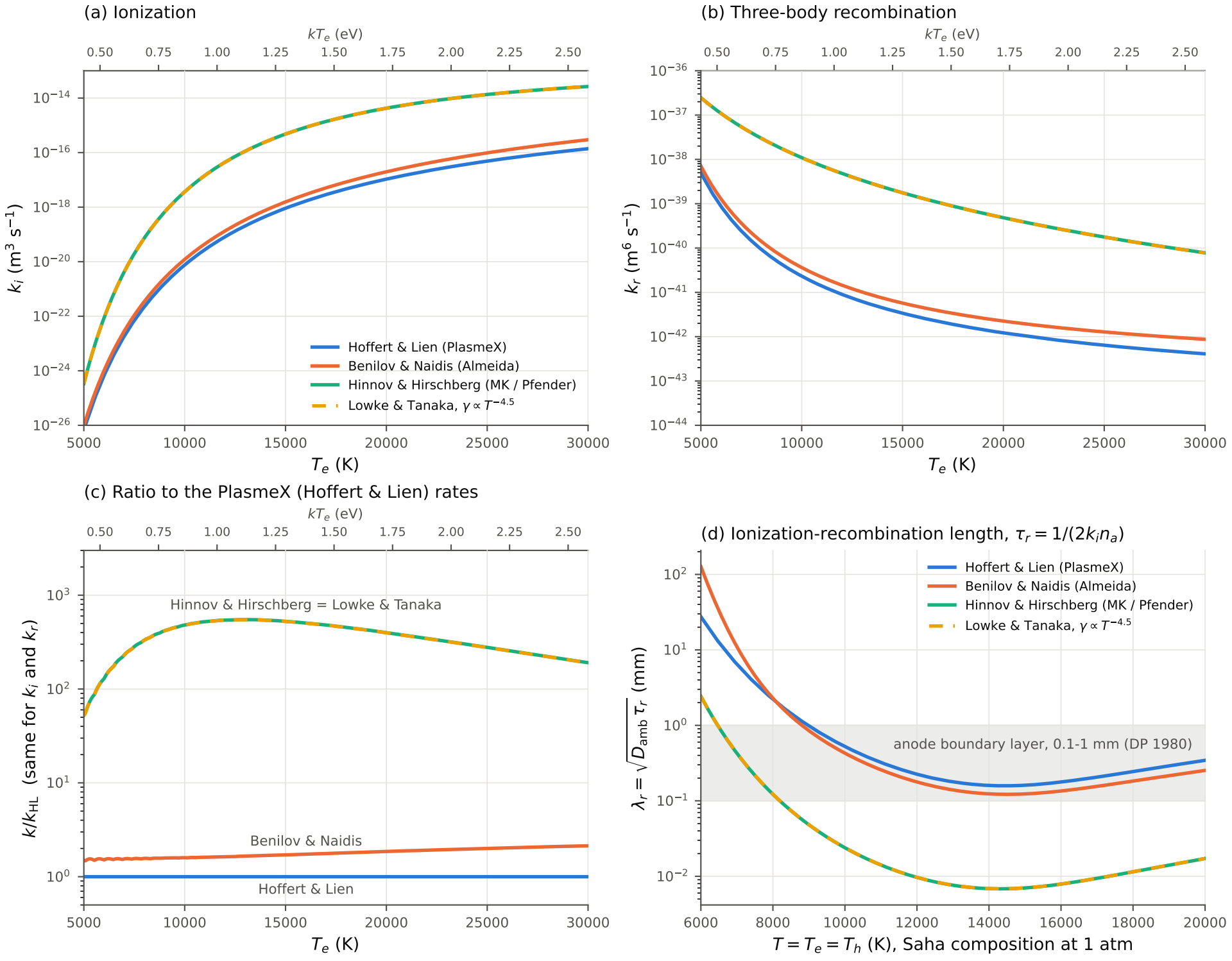
(a) Ionization
kT_e (eV)
0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50
k_i (m^3 s^{-1})
10^{-14} 10^{-16} 10^{-18} 10^{-20} 10^{-22} 10^{-24} 10^{-26}
Hoffert & Lien (PlasmeX)
Benilov & Naidis (Almeida)
Hinnov & Hirschberg (MK / Pfender)
Lowke & Tanaka, $\gamma \propto T^{-4.5}$
5000 10000 15000 20000 25000 30000
T_e (K)
(b) Three-body recombination
kT_e (eV)
0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50
k_r (m^6 s^{-1})
10^{-36} 10^{-37} 10^{-38} 10^{-39} 10^{-40} 10^{-41} 10^{-42} 10^{-43} 10^{-44}
5000 10000 15000 20000 25000 30000
T_e (K)
(c) Ratio to the PlasmeX (Hoffert & Lien) rates
kT_e (eV)
0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50
k/k_{HL} (same for k_i and k_r)
10^3 10^2 10^1 10^0
Hinnov & Hirschberg = Lowke & Tanaka
Benilov & Naidis
Hoffert & Lien
5000 10000 15000 20000 25000 30000
T_e (K)
(d) Ionization-recombination length, $\tau_r = 1/(2k_i n_a)$
Hoffert & Lien (PlasmeX)
Benilov & Naidis (Almeida)
Hinnov & Hirschberg (MK / Pfender)
Lowke & Tanaka, $\gamma \propto T^{-4.5}$
$\lambda_r = \sqrt{D_{amb}\,\tau_r}$ (mm)
10^2 10^1 10^0 10^{-1} 10^{-2}
anode boundary layer, 0.1-1 mm (DP 1980)
6000 8000 10000 12000 14000 16000 18000 20000
$T = T_e = T_h$ (K), Saha composition at 1 atm